

LSTMs and GRUs

DATA 443/643 · Week 2, note 2.3

After CMU 11-785 L14: a cell that can copy, plus input-dependent gates.

Lecture plan

- 1 Why a tanh overwrite cannot remember
- 2 Three LSTM gates and the cell highway
- 3 Copy regime: forget open, input closed
- 4 GRU: two gates, hidden state is memory
- 5 Bidirectional is illegal for next-token LMs

The hidden state is constantly rewritten

Vanilla RNN

Every step: tanh of a new mix. Memory length is an eigenvalue accident.

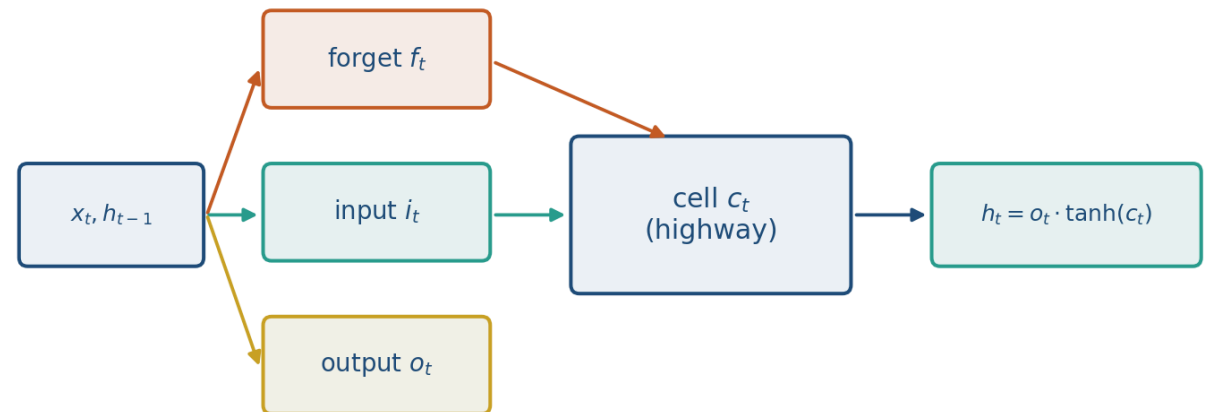
Gated cell

A wire that can copy, plus learned switches that say when to write or erase.

LSTM: a highway plus three gates

- Forget, input, output. Each in (0, 1).
- The cell is mostly multiply and add.
- Hochreiter and Schmidhuber 1997; forget gate Gers 2000.

LSTM: a cell highway plus learned gates



The cell update you will write

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$$

Then $h_t = o_t \odot \tanh(c_t)$.

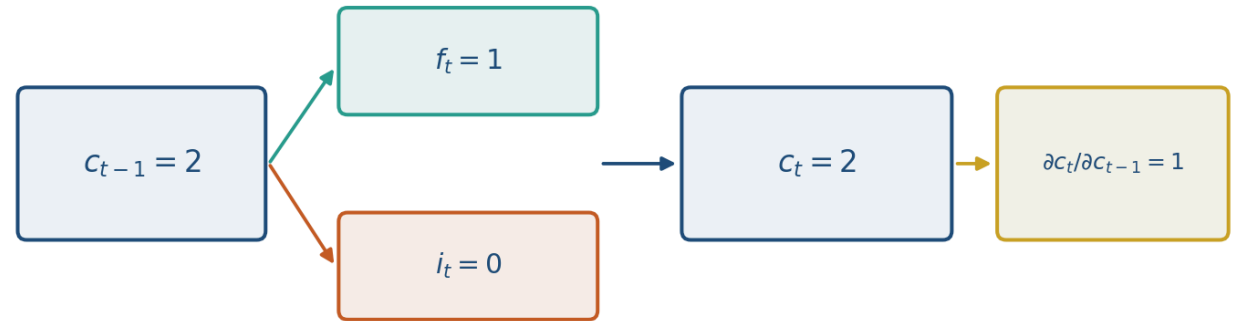
If $f_t \approx 1$ and $i_t \approx 0$, the cell copies.

Along that path,
 $\partial c_t / \partial c_{t-1} = f_t$.

Copy regime

- $c_{t-1} = 2, f = 1, i = 0$ gives $c_t = 2$.
- Ten copies: gradient $1^{10} = 1$.
- Learnable, not guaranteed. About 100 steps, not infinity.

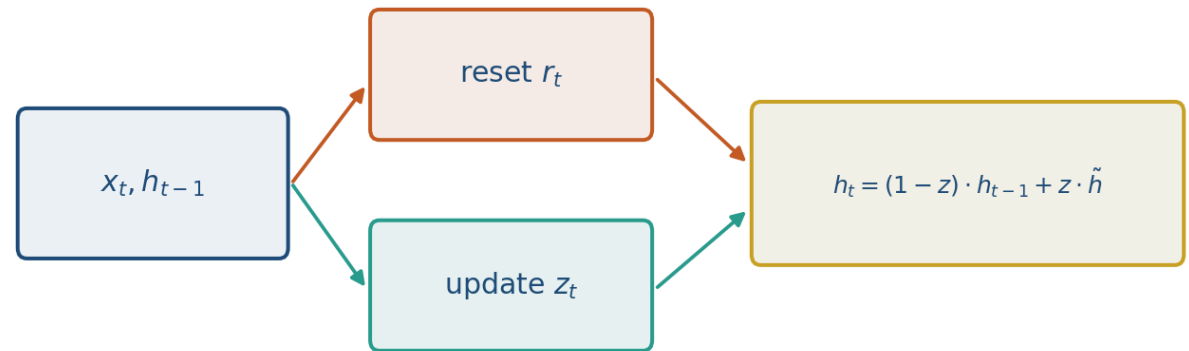
Copy regime: forget open, input closed. The cell is a wire.



GRU: two gates, no extra cell

- Reset and update. The hidden state is the memory.
- $z \approx 0$ copies. $z \approx 1$ replaces.
- Fewer parameters. Still a loop.

GRU: two gates, no separate cell state



When you would still pick a gate

LSTM / GRU

Speech, time series, a small sequential baseline. Lab 2.

Transformer (Week 3)

Long text, GPU parallelism, path length one. The default for this course's projects.

What to take from this lecture

Write it

$c_t = \text{forget} * \text{old} + \text{input} * \text{new}.$

Copy it

$f=1, i=0$ is a wire. Bidirectional
peeks at the future.

Next week

Attention stops using this chain.
Lab 2 is why.